

Nirma University
Institute of Technology
Semester End Examination (IR), January - 2023
Master of Computer Applications, Semester - I
6CS104 Database Management System

Roll /
Exam No.

Supervisor's
initial with date

Time: 3 Hours

Max. Marks: 100

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.
5. Section wise separate answer book is to be used.

SECTION-I

Q-1. Answer the following. [18]

- (A) What do you understand by the database management system? (6)
CO1, BL1 What is the primary goal of a database management system?
Discuss referential integrity with an example.
- (B) Briefly discuss various pros and cons of the file-processing (6)
CO1, BL1 system.
- (C) Why SQL key is important in the database management (6)
CO3, BL2 system? Discuss super key, and candidate key by taking an
appropriate example.

Q-2. Answer the following. [18]

- (A) Discuss the various roles and functions of the database (9)
CO4, BL3 administrator (DBA) in detail.
- (B) Write SQL for creating a table corresponding to the schema as (9)
CO3, BL4 mentioned below:
department (dept name, building, budget)
course (course id, title, dept name, credits)
instructor (ID, name, dept name, salary)
section (course id, sec id, semester, year, building, room
number, time slot id)
teaches (ID, course id, sec id, semester, year)
- Make any reasonable assumptions about data types and
primary key and reference key.

Q-3. Answer the following. [14]

- (A) Discuss the first normal form(1NF), second normal form(2NF), (7)
CO2, BL3 and third normal form(3NF) with appropriate examples.

- (B) Discuss the entity-relationship model. Also, explain the various cardinality with an appropriate example. (7)
CO1, BL2

SECTION-II

Q-4. Answer the following. [18]

- (A) Explain hashing, multilevel index, sparse index, and dense index with appropriate examples. (6)
CO2, BL2
(B) Write SQL for creating a view, updating a view, and dropping the view. Assume and mention the necessary data if required. (6)
CO3, BL4
(C) Write SQL queries to implement the following. Assume and mention the necessary data if required. (6)
CO3, BL5
- I. Inner join
 - II. Left join
 - III. Right join
 - IV. Full outer join
 - V. Having clause
 - VI. Exist clause

Q-5. Answer the following. [18]

- (A) What do you understand by context area? Briefly discuss the attributes of the implicit cursor. Also, write a PL SQL program that updates the table and increases the salary of each customer by 500 and use the SQL%ROWCOUNT attribute to determine the number of rows affected. Assume and mention the necessary data if required. (9)
CO4, BL6
(B) What is the objective of creating a trigger? Write a PL SQL program to create a row-level trigger for the CUSTOMERS table that would fire for INSERT or UPDATE or DELETE operations performed on the CUSTOMERS table. This trigger will display the salary difference between the old values and new values. Assume and mention the necessary data if required. (9)
CO4, BL6

Q-6. Answer the following. [14]

- (A) Write a PL SQL program to demonstrate a user-defined exception. assume and mention the necessary data if required. (7)
CO4, BL6
(B) What do you understand by transaction? Discuss ACID properties in order to ensure accuracy, completeness, and data integrity. (7)
CO3, BL5